Introduction to Photogrammetry

Lecture 07

Image Motion and Image Quality in Aerial Photography

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In Today's Class

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2. Factors Influencing Image Motion
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4. Atmospheric factors Affecting Image Quality

1. Introduction

- High-quality images are essential for accurate photogrammetric measurements.
- Poor image quality makes it difficult to identify objects and measure distances.
- Blurred images reduce the reliability of mapping results.
- Therefore, controlling factors that affect image quality is very important.

What is Image Motion

- Image motion refers to the movement of the image across the film during exposure.
- This movement occurs because the aircraft continues to move while the photograph is being taken.
- As a result, the image of a ground object may shift slightly on the film surface.
- This shift can cause the photograph to appear blurred.

- Imagine taking a photograph from a moving vehicle.
- If the camera shutter remains open for too long, the image becomes blurred.
- A similar situation occurs in aerial photography.
- The aircraft is moving rapidly while the photograph is taken.

2. Factors Influencing Image Motion

I. Aircraft Movement
II. Exposure Time
III. Effect of Flying Height
IV. Effect of Focal Length
V. Aircraft Vibrations

I. Aircraft Movement

- Aircraft used in aerial photography usually fly at high speeds.
- Typical survey aircraft may travel at **200–300 km per hour**.
- Even during a short exposure time, the aircraft moves forward.
- This forward movement causes image displacement.

- The most common cause of image motion is the **forward motion of the aircraft**.
- As the aircraft moves, the image of ground objects moves across the film.
- This movement occurs in the direction of flight.
- The faster the aircraft moves, the greater the image motion.

II. Exposure Time

- Exposure time refers to the duration for which the shutter remains open.
- During this time, light enters the camera and forms the image.
- If the exposure time is too long, image motion increases.
- Therefore, aerial cameras use **very short exposure times**.

III. Effect of Flying Height

- Flying height also affects image motion.
- At lower altitudes, ground features appear larger on the photograph.
- This makes image motion more noticeable.
- At higher altitudes, the effect of motion may be slightly reduced.

IV. Effect of Focal Length

- The focal length of the camera lens also influences image motion.
- Cameras with longer focal lengths produce larger images.
- Larger images make motion effects more noticeable.
- Therefore, focal length must be considered when planning aerial surveys.

V. Aircraft Vibrations

- Another cause of image motion is **aircraft vibration**.
- Engines and air turbulence can cause small vibrations.
- These vibrations can affect the stability of the camera.
- As a result, images may appear slightly blurred.

- To reduce vibration effects, aerial cameras are mounted on **special stabilized platforms**.
- These mounts absorb shocks and vibrations.
- They help maintain a steady camera position.
- Stable mounting improves image quality.

3. Image Motion Compensation

- Some aerial cameras include **image motion compensation systems**.
- These systems move the film slightly during exposure.
- The movement is in the opposite direction of aircraft motion.
- This helps cancel the effect of image displacement.

- During exposure, the aircraft moves forward.
- At the same time, the film moves slightly inside the camera.
- The movement of the film matches the motion of the image.
- As a result, the image remains sharp on the film surface.

Importance of Short Exposure Time

- One of the easiest ways to reduce image motion is to use **short exposure times**.
- Short exposures reduce the time during which motion occurs.
- Modern aerial cameras use very fast shutters.
- This helps produce sharper images.

Limitations of Motion Compensation

- Image motion compensation mainly corrects **forward motion**.
- It cannot fully correct random vibrations.
- Sudden aircraft movements may still affect image quality.
- Therefore, careful flight planning is important.

4. Atmospheric factors Affecting Image Quality

- In aerial photogrammetry, atmospheric factors significantly degrade image quality by reducing contrast, altering colors, and introducing spatial distortions.
- These effects occur as light travels from the sun to the earth and then back up to the camera through layers of the atmosphere.
- Some factors are;
    i. Sun Illumination
    ii. Weather Conditions

i. Sun Illumination

- Good sunlight is important for aerial photography.
- Proper illumination improves image contrast.
- Low sun angles may create long shadows.
- Therefore, aerial photography is usually conducted during **midday hours**.
- Low sun angles (early morning/late afternoon) create long, pronounced shadows that obscure ground features and reduce overall contrast.

ii. Weather Conditions

- Weather conditions also influence image quality.
- Haze, clouds, and dust can reduce visibility.
- Poor atmospheric conditions reduce image clarity.
- Aerial photography is usually conducted in **clear weather**.
- partly cloudy conditions are problematic because shifting shadows create inconsistent exposures across the survey area, making image stitching (orthomosaic creation) difficult.

Flight Planning

- Careful flight planning helps reduce image quality problems.
- Flight altitude and speed are carefully selected.
- Camera settings are adjusted before the mission.
- Good planning ensures high-quality aerial photographs.

Thank you!